

Recommendation Systems for Personalised Content Discovery

Problem Overview

Enhancing User Retention

Netflix manages 480,000 users and 17,000+ movies. Without personalization, users face **Choice Paralysis**, leading to platform churn. Given a platform with thousands of movies and millions of users, how can we accurately predict which unseen movies a specific user will enjoy, thereby improving user engagement and retention.

 **Scale:** Processing 100 Million ratings efficiently.

 **Insight:** The "Long Tail" contains the movies users actually love, but they are buried under blockbusters.

 **Goal:** Predict unseen ratings to surface hidden gems.

APPROACH: SVD



Decomposition

Breaking a massive sparse matrix into two dense, low-rank matrices (User Taste & Item Features).



Latent Factors

Discovering 100 hidden "traits" (e.g., genre, mood, era) purely from mathematical rating patterns.



Optimization

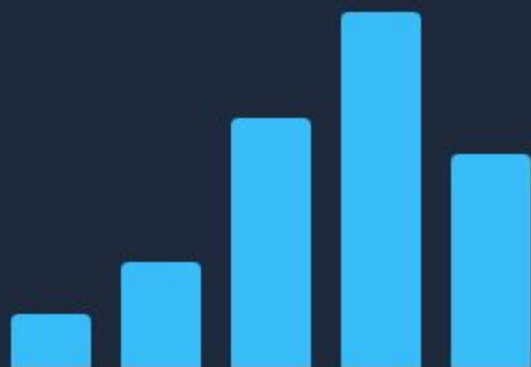
Using 20 epochs of Stochastic Gradient Descent (SGD) to minimize prediction error globally.

Key Insights from EDA



Positivity Bias

Average rating centers at **3.6 stars**. Users rarely rate movies they hate (1s and 2s are outliers), necessitating biased-aware modeling.



The Long Tail

10% of movies receive 90% of all ratings. Our model must prevent "Popularity Gravity" from drowning out niche discoveries.

Minimum rating filters is applied to stabilize the latent space.



User Stratification

Users range from "1-star only" watchers to "5-star only" fans. Without **Normalization**, recommendations become skewed by individual baseline habits.

EXPERIMENTAL RESULTS

Metric	Model A: Funk SVD	Model B: Item-KNN
Prediction Error (RMSE)	0.9	0.9266
Ranking Quality (MAP@10)	0.8177	0.8002
Training Time	154.93 s	152.60 s
Inference Speed	10.72 s	894.18 s

Conclusion: Funk SVD is the only viable production candidate due to its lightning-fast inference and superior ranking.

Top-6 Recommendations Demonstration

User 30878 Results

The West Wing Season 3(2001) 5 ★

The West Wing Season 1(1999) 5 ★

The West Wing Season 4(2002) 5 ★

The West Wing Season 2(1999) 5 ★

Dead Like me Season 2(2004) 4.96 ★

Six Feet under: Season 1 (2001) 4.95 ★

Success Case: Niche Discovery

User 30878 History:

Spitfire Grill 1996

Gross Anatomy 1989

Recommendations:

1. The West wing season 2 5 ★ (1999)

2. The West wing season 1 5 ★ (1999)

The model identified the "90s Fantasy" latent factor without any metadata tags.



Failure Case: Popularity Bias

Problem: Global Bias

For some users, globally popular blockbusters overpower personal latent factors.

Documentary Fan

Rated: Microcosmos (1996)

Rec: Lord of the Rings

The high popularity bias pushed LOTR to the top, ignoring the documentary preference.

Observations:

Bias Dominance: High-rating blockbusters like "Shawshank Redemption" have such high $\$b_i$ that they bubble up even for users who only watch Indie

Cold Item Failure: Movies with <50 ratings cannot form a distinct latent vector, making them invisible to the system.